Probability Notes

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1 Events and their Probabilities

Definition 1.1 (Sample Space)**.** The set of all possible outcomes of an experiment is called the **sample space** and is denoted by Ω. An **event** is a subset of the sample space.

Example 1.1. Consider an experiment of flipping of a coin three times. The sample space is

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Let us define an event A indicating that majority of coins shows up head. Therefore

$$A = \{HHH, HHT, HTH, THH\}.$$

Example 1.2. Consider an experiment of rolling two dice. The sample space is

$$\begin{aligned} S = \{&(1,1),(1,2),\ldots,(1,6),\\ &(2,1),(2,2),\ldots,(2,6),\\ &\vdots \quad \vdots \quad \ldots \quad \vdots\\ &(6,1),(6,2),\ldots,(6,6)\}. \end{aligned}$$

Let us define an event A indicating that sum of the faces showing is 7. Therefore

$$A = \{(1,6),(2,5),(5,2),(3,4),(4,3),(6,1)\}.$$

There are three ways to indicate an experimenter's sample space:

1. If the number of possible outcomes is small then listing can be done.
2. In some cases structure can be shown.
3. State a mathematical formula that sample outcomes must satisfy.
   For example a programmer is running a subroutine that solves a general quadratic equation $ax^2 + bx + c = 0$. Experiment consist of choosing the values for three coefficients a, b and c. In this case $S = \{(a,b,c) : \ \infty < a < \infty, \ \infty < b < \infty, \ \infty < c < \infty, \}$. If the event A: Equation has two equal roots, then $A = \{(a,b,c) : b^2 - 4ac = 0\}$.

There are three very different ways of defining probability of an event.

1. **Classical Definition** : Imagine an experiment, or game, having n possible outcomes-and suppose that those outcomes are equally likely. If some event A were satisfied by m of those n then
   $$P(A) = \frac{m}{n}.$$
   Limitations of this definition are
   (i) What if outcomes of the experiments are not equally likely? or
   (ii) the number of outcomes is not finite.
2. **Empirical Definition**: Consider a sample space, S, and any event, A, defined on S. If experiment were performed one time, either A or A^c would be the outcome. If it were performed n times, and let us suppose that m times A has occurred $0 \leq m \leq n$. Now repeat the process an infinite number of times. As n gets large, the ratio m/n will fluctuate less and less, and it will coverage to some number. This number is called $P(A)$, that is,
   $$P(A) = \lim_{n\to\infty} \frac{m}{n}.$$
   It is shown in the following figure:
   For example, if a fair coin were to be tossed repeatedly with A being the event "Head appears". It is reasonable to expect that $P(A) = \lim_{n\to\infty} \frac{m}{n}$ would be equal to $1/2$.

There are some limitations of this definition e.g., (i) how large n must be to give a good approximation for $P(A)$? (ii) what if the coin is biased? (iii) repeating the experiment under identical conditions for an infinite number of times is physically impossible.

Before we discuss next definition, it is observed that, probability is not defined as a ratio, or as a limit to the ratio, but simply as a function that behaved in accordance with a prescribed set of axioms.

3. **Axiomatic Definition**: If S has a finite number of members, the necessary and sufficient set of axioms for P are:
   Axiom 1: Let A be any event defined over S. Then $P(A) \geq 0$.
   Axiom 2: $P(S) = 1$
   Axiom 3: If A and B be any two mutually exclusive (disjoint) events defined over S. Then

$$P(A \cup B) = P(A) + P(B). \tag{1}$$

   When S has an infinite number of members, a fourth axioms is needed:
   Axiom 4: Let $A_1, A_2, \ldots,$ be events defined over S. If $A_i \cap B_j = \Phi$ for each $i \neq j$, then

$$P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i). \tag{2}$$

The axiomatic definition of probability can be described/generalised from the perspective of measure as below.

Definition 1.2 (Field)**.** Any collection $\mathscr{F}$ of subsets of Ω which satisfies the following three conditions is a **field**:

1. if $A, B \in \mathscr{F}$ then $A \cup B \in \mathscr{F}$ and $A \cap B \in \mathscr{F}$,
2. if $A \in \mathscr{F}$ then $A^c \in \mathscr{F}$,
3. the empty set ϕ belongs to $\mathscr{F}$.

Definition 1.3 ($\sigma-$ field)**.** A collection $\mathscr{F}$ of subsets of Ω is called a $\sigma-$ field if it satisfies the following:

1. $\phi \in \mathscr{F}$,
2. if $A_1, A_2, \ldots \in \mathscr{F}$ then $\cup_{i=1}^{\infty} \in \mathscr{F}$,
3. if $A \in \mathscr{F}$ then $A^c \in \mathscr{F}$.

Definition 1.4 (Probability Space)**.** A **probability measure** on $(\Omega, \mathscr{F})$ is a function $\mathbb{P} : \mathscr{F} \to [0, 1]$ satisfying

1. $\mathbb{P}(\phi) = 0, \mathbb{P}(\Omega) = 1$,
2. if $A_1, A_2, \ldots$ is a collection of disjoint events of $\mathscr{F}$ such that $A_i \cap A_j = \phi \ \forall i \neq j, then$

$$\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

$(\Omega, \mathscr{F}, \mathbb{P})$ is called the **probability space**.

Theorem 1.1. If A and B are two disjoint events in Ω, then

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$,
2. if $B \supseteq A$, then $\mathbb{P}(B) \geq \mathbb{P}(A)$,

3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$,

Proof: (1) From definition 1.4, $\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(A \cup A^c) = \mathbb{P}(\Omega) = 1$, Thus, $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$.
(2) $\mathbb{P}(B) = \mathbb{P}(A \cup (B \backslash A)) = \mathbb{P}(A) + \mathbb{P}(B \backslash A) \geq \mathbb{P}(A)$, since $\mathbb{P}(B \backslash A) \geq 0$.
(3) We know that A can be written as a disjoint union of $A \backslash B, B \backslash A,$ and $A \cup B$, then

$$\begin{aligned}
\mathbb{P}(A \cup B) &= \mathbb{P}(A \backslash B) + \mathbb{P}(B \backslash A) + \mathbb{P}(A \cap B) \\
&= \mathbb{P}(A \backslash (A \cap B)) + \mathbb{P}(B \backslash (A \cup B)) + \mathbb{P}(A \cap B) \\
&= \mathbb{P}(A) - \mathbb{P}(A \cup B) + \mathbb{P}(B) - \mathbb{P}(A \cup B) + \mathbb{P}(A \cup B) \\
&= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).
\end{aligned}$$

Theorem 1.2 (**Principle of Inclusion and Exclusion**)**.** For events $A_1, A_2, \ldots, A_n$ in a probability space $(\Omega, \mathscr{F}, \mathbb{P})$,

$$\begin{aligned}
\mathbb{P}(\cup_{i=1}^n) A_i = \sum_{i=1}^n \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} \mathbb{P}(A_i \cap A_j \cap A_k) + \cdots \\
+ (-1)^{n-1} \sum_{1 \leq i < \ldots j \leq n} \mathbb{P}(\cap_{i=1}^n A_i).
\end{aligned}$$

Proof: Proof by induction on n.

Theorem 1.3 (**Boole's Inequality**)**.** For events $A_1, A_2, \ldots, A_n$ in a probability space $(\Omega, \mathscr{F}, \mathbb{P})$,

$$\mathbb{P}(\cup_{i=1}^n A_i) \leq \sum_{i=1}^n \mathbb{P}(A_i), \quad \mathbb{P}(\cap_{i=1}^n A_i) \geq 1 - \sum_{i=1}^n \mathbb{P}(A_i^c).$$

Proof: Proof by induction on n.

Definition 1.5 (Independent Events)**.** Events A and B are called **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

More generally, a family of events $\{A_i : i \in I\}$ are independent if

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$$

for all finite subsets J of I.

Theorem 1.4. If A and B are independent, then the following pair of events are also independent:

(a) A, B^c

(b) A^c, B

(c) A^c, B^c

Proof: A and B are independent implies $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Thus,

$$\begin{aligned}
(a)\ \mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A)[1 - \mathbb{P}(B)] = \mathbb{P}(A)\mathbb{P}(B^c), \\
(b)\ \mathbb{P}(A^c \cap B) &= \mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(B)[1 - \mathbb{P}(A)] = \mathbb{P}(A^c)\mathbb{P}(B), \\
(c)\ \mathbb{P}(A^c \cap B^c) &= \mathbb{P}((A \cup B)^c) = 1 - \mathbb{P}(A \cup B) = 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A \cap B) \\
&= \mathbb{P}(A^c) - \mathbb{P}(B) + \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A^c) - \mathbb{P}(B)\mathbb{P}(A^c) \\
&= \mathbb{P}(A^c)[1 - \mathbb{P}(B)] = \mathbb{P}(A^c)\mathbb{P}(B^c).
\end{aligned}$$

Theorem 1.5. (a)Let $A_1, A_2, \ldots$ be an increasing sequence of events, so that $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$, and write A for their limit:

$$A = \bigcup_{i=1}^{\infty} A_i = \lim_{i\to\infty} A_i.$$

Then $\mathbb{P}(A) = \lim_{i\to\infty} \mathbb{P}(A_i)$.
(b) Let $B_1, B_2, \ldots$ be a decreasing sequence of events, so that $B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots$, and write B for their limit:

$$B = \bigcap_{i=1}^{\infty} B_i = \lim_{i\to\infty} B_i.$$

Then $\mathbb{P}(B) = \lim_{i\to\infty} \mathbb{P}(B_i)$.

Proof: (a) We can write $A = A_1 \cup A_2 \cup A_3 \cup \ldots = A_1 \cup (A_2 \backslash A_1) \cup (A_3 \backslash A2) \cap \ldots$. Thus,

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A_1) + \mathbb{P}(A_2\backslash A_1) + \mathbb{P}(A_3\backslash A_2) + \ldots \\ &= \mathbb{P}(A_1) + \mathbb{P}(A_2) - \mathbb{P}(A_1) + \mathbb{P}(A_3) - \mathbb{P}(A_2) + \ldots \\ &= \lim_{i\to\infty} \mathbb{P}(A_i).\end{aligned}$$

(b) Since $B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots$, then $B_1^c \subseteq B_2^c \subseteq B_3^c \subseteq \cdots$. Thus, by part (1) of this theorem

$$\mathbb{P}(B) = 1 - \mathbb{P}(B^c) = 1 - \lim_{i\to\infty} B_i^c = 1 - \lim_{i\to\infty}(1 - B_i) = \lim_{i\to\infty} B_i.$$

First- Borel Cantelli Lemma: If $(A_n)_n$ is a sequence of events such that $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(limsup_n A_n) = 0$.
Proof: Let $B_m = \cup_{n\geq m} A_n$ then $B_{m+1} \subset B_m \ \forall\ m$, and hence

$$\begin{aligned}\mathbb{P}(limsup_n A_n) &= \mathbb{P}(\cap_{m\geq 1}(\cup_{n\geq m} A_n)) = \mathbb{P}(\cap_{m\geq 1} B_m) = \lim_{m\to\infty} \mathbb{P}(B_m) \\ &= \lim_{m\to\infty} \mathbb{P}(\cup_{n\geq m} A_n) \leq \lim_{m\to\infty} (\mathbb{P}(A_m) + \mathbb{P}(A_{m+1}) + \ldots) \\ &= \lim_{m\to\infty} \left(\sum_{n=1}^{\infty} \mathbb{P}(A_n) - \sum_{n=1}^{m-1} \mathbb{P}(A_n) \right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n) - \lim_{m\to\infty} \left(\sum_{n=1}^{m-1} \mathbb{P}(A_n) \right) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(A_n) - \sum_{n=1}^{\infty} \mathbb{P}(A_n) = 0.\end{aligned}$$

The second last statement is justified by the fact that $\sum_{n=1}^{\infty} \mathbb{P}(A_n)$ converges.

Second- Borel Cantelli Lemma: If the sequence of events $(A_n)_n$ is independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(limsup_n(A_n)) = 1$.
Proof: Since $\mathbb{P}(A_n) \geq 0$ implies $1 - \mathbb{P}(A_n) \leq e^{-\mathbb{P}(A_n)}$, for $k \leq m$,

$$\mathbb{P}(\cap_{n=k}^{m} A_n^c) = \prod_{n=k}^{m} \mathbb{P}(A_n^c) = \prod_{n=k}^{m} (1 - \mathbb{P}(A_n)) \leq \prod_{n=k}^{m} e^{-\mathbb{P}(A_n)} = e^{-\sum_{n=k}^{m} \mathbb{P}(A_n)},$$

$$\begin{aligned}1 - \mathbb{P}(\cup_{n=k}^{\infty} A_n) &= 1 - \lim_{m\to\infty} \mathbb{P}(\cup_{n=k}^{m} A_n) = \lim_{m\to\infty} (1 - \mathbb{P}(\cup_{n=k}^{m} A_n)) \\ &= \lim_{m\to\infty} \mathbb{P}((\cup_{n=k}^{m} A_n)^c) \lim_{m\to\infty} \mathbb{P}(\cap_{n=k}^{\infty} A_n^c) \\ &\leq \lim_{m\to\infty} e^{-\sum_{n=k}^{m} \mathbb{P}(A_n)} = e^{-\lim_{m\to\infty} \sum_{n=k}^{m} \mathbb{P}(A_n)} \\ &= e^{-\infty} = 0.\end{aligned}$$

Thus, $\mathbb{P}(\cup_{n=k}^{\infty} A_n) = 1 \ \forall\ k$, and hence

$$1 = \lim_{k\to\infty} \mathbb{P}(\cup_{n=k}^{\infty} A_n) = \mathbb{P}(\cap_{k=1}^{\infty}(\cup_{n=k}^{\infty} A_n)) = \mathbb{P}(limsup_n(A_n)).$$

Definition 1.6 (Conditional Probability)**.** If $\mathbb{P}(B) > 0$ then the **conditional probability** that A occurs given that B occurs is defined to be

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Theorem 1.6. (a) For any events A and B such that $0 < \mathbb{P}(B) < 1$,

$$\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c).$$

(b) More generally, let $B_1, B_2, \ldots, B_n$ be a partition of A such that $\mathbb{P}(B_i) > 0$ for all i. Then

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

Proof: (a) We can write A as a disjoint union of $A \cap B$ and $A \cap B^c$. Then using the definition of conditional probability

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \\ &= \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c).\end{aligned}$$

(b) We can write A as a disjoint union of $A \cap B_i$ where $i \in \{1, 2, \ldots, n\}$. Then using the definition of conditional probability

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap B_1) + \mathbb{P}(A \cap B_1) + \ldots + \mathbb{P}(A \cap B_n) \\ &= \mathbb{P}(A|B_1)\mathbb{P}(B_1) + \mathbb{P}(A|B_2)\mathbb{P}(B_2) + \ldots + \mathbb{P}(A|B_n)\mathbb{P}(B_n).\end{aligned}$$

Theorem 1.7 (**Bayes Theorem**)**.** Let $A_1, A_2, \ldots, A_n$ be a partition of Ω each A_i having positive probability, then for any event B,

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}.$$

Proof: Using the definition of conditional probability and last theorem, we can write

$$\begin{aligned}\mathbb{P}(A_j|B) &= \frac{\mathbb{P}(A_j \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A_j \cap B)}{\mathbb{P}((B \cap A_1) \cup (B \cap A_2) \cup \cdots \cup (B \cap A_n))} \\ &= \frac{\mathbb{P}(A_j \cap B)}{\sum_{i=1}^{n} \mathbb{P}(B \cap A_i)} = \frac{\mathbb{P}(B|A_j)P(A_j)}{\sum_{i=1}^{n} \mathbb{P}(B|A_j)P(A_j)}.\end{aligned}$$

Example 1.3. A rare disease affects one person in 10^5. A test for the disease shows $+ve$ with probability $99/100$ when applied to an ill person, and with probability $1/100$ when applied to an healthy person. What is the probability that you have the disease given that the test shows $+ve$?

$$\begin{aligned}\mathbb{P}(ill| + ve) &= \frac{\mathbb{P}(+ve|ill)\mathbb{P}(ill)}{\mathbb{P}(+ve|ill)\mathbb{P}(ill) + \mathbb{P}(+ve|healthy)\mathbb{P}(healthy)} \\ &= \frac{\frac{99}{100} \times \frac{1}{10^5}}{\frac{99}{100} \times \frac{1}{10^5} + \frac{1}{100} \times \frac{10^5-1}{10^5}} = \frac{99}{100098}.\end{aligned}$$

2 Random Variables and their Distribution

Definition 2.1 (Random Variable)**.** A **random variable** is a function $X : \Omega \to \mathbb{R}$ with the property that $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathscr{F}$ for each $x \in \mathbb{R}$. Such a function is said to be $\mathscr{F}-$ measurable.

Definition 2.2 (Distribution Function)**.** The **distribution function** of a random variable X is the function $F : \mathbb{R} \to [0, 1]$ given by

$$F_X(x) = \mathbb{P}(X \leq x); \forall\, x \in \mathbb{R}.$$

Example 2.1. Let $\Omega = HH, HT, TH, TT$ be a sample space of tossing a coin twice where H and T represents head and tail, respectively. Then $X : \Omega \to \mathbb{R}$ defined by $X(\omega) = no.\ of\ heads\ in\ \omega$, is a random variable where

$$X = \begin{cases} 0 & with\ probability\ 1/4 \\ 1 & with\ probability\ 1/2 \\ 2 & with\ probability\ 1/4. \end{cases}$$

The distribution function of the random variable X is given by

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1/4 & 0 \leq x < 1 \\ 3/4 & 1 \leq x < 2 \\ 1 & x > 2. \end{cases}$$

Theorem 2.1. Let F be the distribution function of a random variable X, then

(a) $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$,

(b) if $x < y$, then $F(x) \leq F(y)$,

(c) F is right-continuous function, that is, $F(x + h) \to F(x)$ as $h \downarrow 0$,

Proof: (a) Let $B_n = \{\omega \in \Omega : X(\omega) \leq -n\}$, then $B_1 \supseteq B_2 \supseteq B_3 \supseteq \ldots$ with $\lim_{n\to\infty} B_n = \phi$. Then

$$\lim_{x\to-\infty} F_X(x) = \lim_{x\to-\infty} \mathbb{P}(X \leq x) = \lim_{n\to\infty} \mathbb{P}(B_n) = \mathbb{P}(\phi) = 0.$$

Let $A_n = \{\omega \in \Omega : X(\omega) \leq n\}$, then $A_1 \subseteq A_2 \subseteq A_3 \subseteq \ldots$ with $\lim_{n\to\infty} A_n = \Omega$. Then

$$\lim_{x\to\infty} F_X(x) = \lim_{x\to\infty} \mathbb{P}(X \leq x) = \lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(\Omega) = 1.$$

(b) For $x < y$, let $A_x = \{\omega \in \Omega : X(\omega) \leq x\}, A_y = \{\omega \in \Omega : X(\omega) \leq y\}$, and $A_{xy} = \{\omega \in \Omega : x < X(\omega) \leq y\}$, then $A_y = A_x \cup A_{xy}$. Since A_x and A_{xy} are disjoint, $\mathbb{P}(A_y) = \mathbb{P}(A_x) + \mathbb{P}(A_{xy})$, i.e.,

$$F(y) = F(x) + \mathbb{P}(x < X \leq y) \geq F(x).$$

(c) For $x \in \mathbb{R}$, let $A_n = \{\omega \in \Omega : X(\omega) \leq x + \frac{1}{n}\}$ and $B_n = \{\omega \in \Omega : X(\omega) \leq x - \frac{1}{n}\}$. Since A_n is a decreasing sequence of events with $\lim_{n\to\infty} A_n = \{\omega \in \Omega : X(\omega) \leq x\}$ and B_n is an increasing sequence of events with $\lim_{n\to\infty} B_n = \{\omega \in \Omega : X(\omega) < x\}$,

$$\begin{aligned} \lim_{h\downarrow 0} F(x + h) &= \lim_{h\downarrow 0} \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x + h\}) = \lim_{n\to\infty} \mathbb{P}(A_n) \\ &= \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = F(x), \end{aligned}$$

and

$$\begin{aligned} \lim_{h\downarrow 0} F(x - h) &= \lim_{h\downarrow 0} \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x - h\}) = \lim_{n\to\infty} \mathbb{P}(B_n) \\ &= \mathbb{P}(\{\omega \in \Omega : X(\omega) < x\}) = F(x^-) \ \ (\neq F(x)). \end{aligned}$$

Thus, the distribution function F is right-continuous but not left-continuous.
In fact, a function F is the distribution function of some random variable if and only if it satisfies the properties of the above theorem.

Theorem 2.2. Let F be a distribution function of a random variable X, then:

(a) $\mathbb{P}(X > x) = 1 - F(x)$,

(b) $\mathbb{P}(x < X \leq y) = F(y) - F(x)$,

(c) $\mathbb{P}(X = x) = F(x) - \lim_{y \uparrow x} F(y)$.

Proof: (a) $\mathbb{P}(X \leq x) + \mathbb{P}(X > x) = 1$, implies $\mathbb{P}(X > x) = 1 - \mathbb{P}(X \leq x) = 1 - F(x)$.
(b) For $x < y$,

$$\begin{aligned}
\mathbb{P}(x < X \leq y) &= \mathbb{P}(\{\omega \in \Omega : x < X(\omega) \leq y\}) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq y\} \cap \{\omega \in \Omega : x < X(\omega)\}) \\
&= P(\{\omega \in \Omega : X(\omega) \leq y\}) + P(\{\omega \in \Omega : x < X(\omega)\}) \\
&\quad - P(\{\omega \in \Omega : X(\omega) \leq y\} \cup \{\omega \in \Omega : x < X(\omega)\}) \\
&= F(y) + 1 - F(x) - P(\{\omega :\in \Omega : X(\omega) \in \mathbb{R}\}) = F(y) + 1 - F(x) - 1 \\
&= F(y) - F(x).
\end{aligned}$$

(c) Let $B_n = \{x - 1/n < X \leq x\}$, then B_n is a decreasing sequence with limit $B = \{X = x\}$. Thus,

$$\begin{aligned}
\mathbb{P}(X = x) = \mathbb{P}(B) &= \lim_{n\to\infty} \mathbb{P}(B_n) = \lim_{n\to\infty} \mathbb{P}(\{x - 1/n < X \leq x\}) \\
&= \lim_{n\to\infty} \mathbb{P}(\{X \leq x\} \cap \{X \leq x - 1/n\}^c) \\
&= \lim_{n\to\infty} (\mathbb{P}(\{X \leq x\}) + \mathbb{P}(\{X \leq x - 1/n\}^c) - \mathbb{P}(\{X \leq x\} \cup \{X \leq x - 1/n\}^c)) \\
&= \mathbb{P}(X \leq x) + \lim_{n\to\infty} \mathbb{P}(\{X \leq x - 1/n\}^c) - \lim_{n\to\infty} \mathbb{P}(X \in \mathbb{R}) \\
&= F(x) + 1 - \lim_{n\to\infty} \mathbb{P}(\{X \leq x - 1/n\}) - 1 \\
&= F(x) - \lim_{n\to\infty} F(x - 1/n) = F(x) - \lim_{y\uparrow x} F(y).
\end{aligned}$$

Example 2.2. A random variable X has distribution function F. Find the distribution function of $Y = aX + b$, where $a, b \in \mathbb{R}$.
For $a > 0$, $F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(aX + b \leq y) = \mathbb{P}(X \leq \frac{y-b}{a}) = F(\frac{y-b}{a})$.
For $a < 0$,

$$\begin{aligned}
F_Y(y) = \mathbb{P}(Y \leq y) = F(aX + b \leq y) = \mathbb{P}(aX \leq y - b) &= \mathbb{P}\left(X \geq \frac{y-b}{a}\right) \\
&= 1 - \mathbb{P}\left(X < \frac{y-b}{a}\right) = 1 - \mathbb{P}\left(X \leq \frac{y-b}{a}\right) + \mathbb{P}\left(X = \frac{y-b}{a}\right) \\
&= 1 - F\left(\frac{y-b}{a}\right) + \mathbb{P}\left(X = \frac{y-b}{a}\right).
\end{aligned}$$

For $a = 0$,

$$F_Y(y) = \begin{cases} 0, & if\ y < b, \\ 1, & if\ y \geq b. \end{cases}$$

Example 2.3. Show that if F and G are distribution functions and $0 \leq \lambda \leq 1$, then $\lambda F + (1 - \lambda)G$ is a distribution.
Let $H(x) = \lambda F(x) + (1 - \lambda)G(x)$.
(a) If $x < y$, then $F(x) \leq F(y)$ and $G(x) \leq G(y)$; this implies $H(x) \leq H(y)$.
(b) $\lim_{x\to-\infty} H(x) = \lambda \lim_{x\to-\infty} F(x) - (1 - \lambda) \lim_{x\to-\infty} G(x) = \lambda \times 0 + (1 - \lambda) \times 0 = 0$,
and $\lim_{x\to\infty} H(x) = \lambda \lim_{x\to\infty} F(x) + (1 - \lambda) \lim_{x\to\infty} G(x) = \lambda \times 1 + (1 - \lambda) \times 1 = 1$.
(c) Since F and G are right continuous, $\lambda F + (1 - \lambda)G$ is right continuous by real analysis.
By theorem 2.2, (a), (b), and (c) proves that $\lambda F + (1 - \lambda)G$ is a distribution function.

Definition 2.3 (Discrete Random Variable)**.** A random variable is called **discrete** if it takes values in some countable subset $\{x_1, x_2, \ldots\}$ of $\mathbb{R}$. The discrete random variable has **probability mass function** $f : \mathbb{R} \to [0, 1]$ given by

$$f(x) = \mathbb{P}(X = x)$$

Definition 2.4 (Continuous Random Variable)**.** The random variable X is called **continuous** if its distribution function can be expressed as

$$F(x) = \int_{-\infty}^{x} f(u)du \ \forall \ x \in \mathbb{R},$$

for some integrable function $f : \mathbb{R} \to [0, \infty)]$ called the **probability density function** of X.

The distribution function of these random variables can be written as:

$$F(x) = \begin{cases} \sum_{i=-\infty}^{x} f(i) & \text{if X is discrete r.v.,} \\ \int_{-\infty}^{x} f(u)du & \text{if X is continuous r.v.} \end{cases}$$

Note that $F(\infty) = 1$, in both of the above cases.

Lemma: The probability mass function $f : \mathbb{R} \to [0, 1]$ satisfies:

(a) the set of x such that $f(x) \neq 0$ is countable,

(b) $\sum_i f(x_i) = 1$ where $x_i's$ are values in $\mathbb{R}$ such that $f(x_i) \neq 0$.

Proof: The proof is obvious.

Definition 2.5 (Random Vector)**.** A **random vector** is n random variables $X_1, X_2, \ldots, X_n$ written in vector form as $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$ which takes values in $\mathbb{R}^n$.

For vectors $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ and $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$, we write $\boldsymbol{x} \leq \boldsymbol{y}$ if $x_i \leq y_i$ for each $i = 1, 2, \ldots, n$.

Definition 2.6 (Joint Distribution Function)**.** The **joint distribution function** of a random vector $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$ on the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ is the function $F_{\boldsymbol{X}} : \mathbb{R}^n \to [0, 1]$ given by $F_{\boldsymbol{X}}(\boldsymbol{x}) = \mathbb{P}(\boldsymbol{X} \leq \boldsymbol{x})$ for $\boldsymbol{x} \in \mathbb{R}^n$.

Theorem 2.3. The joint distribution function of the random vector $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$ has the following properties:

(a) $\lim_{\boldsymbol{x} \to -\infty} F_{\boldsymbol{X}}(\boldsymbol{x}) = 0$, $\lim_{\boldsymbol{x} \to \infty} F_{\boldsymbol{X}}(\boldsymbol{x}) = 1$,

(b) if $\boldsymbol{x} \leq \boldsymbol{y}$, then $F_{\boldsymbol{X}}(\boldsymbol{x}) \leq F_{\boldsymbol{X}}(\boldsymbol{y})$,

(c) $F_{\boldsymbol{X}}$ is continuous from above, in that,

$$F_{\boldsymbol{X}}(\boldsymbol{x} + \boldsymbol{h}) \to F_{\boldsymbol{X}}(\boldsymbol{x}) \text{ as } \boldsymbol{h} \downarrow \boldsymbol{0},$$

where $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$.

Proof: Apply Theorem 2.1 to each coordinate of the random vector to get the result.

Let $F_{X,Y}$ be the joint distribution function of random variables X and Y, then the marginal distribution functions of X and Y, F_X and F_Y, can be obtained from their joint distribution function as follows:

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x \ \& \ -\infty < Y < \infty) = \mathbb{P}(X \leq x \ \& \ Y < \infty) \\ &= F_{X,Y}(x, \infty) = \lim_{y \to \infty} F_{X,Y}(x, y). \end{aligned}$$

Similarly, $F_Y(y) = \lim_{x \to \infty} F_{X,Y}(x, y)$.

3 Discrete Random Variable

Recall that, a random variable is called **discrete** if it takes values in some countable subset $\{x_1, x_2, \ldots\}$ of $\mathbb{R}$. The discrete random variable has **probability mass function** $f : \mathbb{R} \to [0, 1]$ given by

$$f(x) = \mathbb{P}(X = x)$$

Definition 3.1 (Independent Discrete Random Variables)**.** Discrete variables X and Y are **independent** if the events $\{X = x\}$ and $\{Y = y\}$ are independent for all x and y, that is,

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y).$$

Definition 3.2 (Expectation)**.** The **mean value**, or **expectation** of a discrete random variable X with mass function f is defined to be

$$\mathbb{E}(X) = \sum_x x f(x)$$

whenever this sum is absolutely convergent.

Theorem 3.1. The expectation operator $\mathbb{E}$ has the following properties:

(a) if $X \geq 0$ then $\mathbb{E}(X) \geq 0$,

(b) if $a, b \in \mathbb{R}$ then $\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$,

(c) $\mathbb{E}(c) = c$ for a constant $c \in \mathbb{R}$.

Proof: (a) If $X \geq 0$, it takes non-negative values only. Also for X taking any value x, $f(x) = \mathbb{P}(X = x) \geq 0$. Thus, $\mathbb{E}(X) = \sum_x x f(x) \geq 0$.

(b)

$$\begin{aligned}
\mathbb{E}(aX + bY) &= \sum_x \sum_y (ax + by) f(x, y) = \Big(\sum_x ax \sum_y f(x, y)\Big) + \Big(\sum_y by \sum_x f(x, y)\Big) \\
&= \sum_x ax f_X(x) + \sum_y by f_Y(y) = a \sum_x x f_X(x) + b \sum_y y f_Y(y) \\
&= a\mathbb{E}(X) + b\mathbb{E}(Y).
\end{aligned}$$

(c) $\mathbb{E}(c) = \sum_x x\mathbb{P}(X = c) = c\mathbb{P}(X = c) = c \cdot 1 = c$.

Theorem 3.2. If X and Y are independent, then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.
Proof: Let $f_X, f_Y, f_{X,Y}$ be the mass function of $X, Y, (X, Y)$, respectively. Since X and Y are independent, $f_{X,Y}(x, y) = f_X(x) f_Y(y)$. And hence,

$$\begin{aligned}
\mathbb{E}(XY) &= \sum_x \sum_y xy f_{X,Y}(x, y) = \sum_x \sum_y xy f_X(x) f_Y(y) \\
&= \Big(\sum_x x f_X(x)\Big)\Big(\sum_y y f(y)\Big) = \mathbb{E}(X)\mathbb{E}(Y).
\end{aligned}$$

Definition 3.3 (Uncorrelated Random Variable)**.** X and Y are **uncorrelated** random variables if

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y).$$

Theorem 3.3. If the random variable X has a mass function, and $g : \mathbb{R} \to \mathbb{R}$, then

$$\mathbb{E}(g(X)) = \sum_x g(x) f(x),$$

whenever this sum is absolutely convergent.

Proof: Let $Y = g(X)$, then

$$\begin{aligned}\mathbb{E}[g(X)] = \mathbb{E}[Y] &= \sum_y y f_Y(y) = \sum_y y\mathbb{P}(Y = y) = \sum_y y\mathbb{P}(g(X) = y) \\ &= \sum_y y \sum_{x:g(x)=y} \mathbb{P}(X = x) = \sum_y y \sum_{x:g(x)=y} f_X(x) = \sum_x g(x) f_X(x).\end{aligned}$$

Definition 3.4 (kth moment)**.** If k is a positive integer, the kth **moment** m_k of X is defined to be

$$m_k = \mathbb{E}(X^k).$$

The kth **central moment** σ_k is

$$\sigma_k = \mathbb{E}((X - m_1)^k).$$

Definition 3.5 (Variance)**.** The **variance** of a discrete random variable X is defined to be

$$Var(X) = \mathbb{E}[(X - \mathbb{E}(X))^2].$$

Variance of a random variable X can also be written as:

$$\begin{aligned}Var(X) &= \mathbb{E}[(X - m_1)^2] = \mathbb{E}[X^2 - 2m_1X + m_1^2] = \mathbb{E}(X^2) - 2m_1\mathbb{E}(X) + \mathbb{E}(m_1^2) \\ &= \mathbb{E}(X^2) - 2m_1^2 + m_1^2 = m_2 - m_1^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2.\end{aligned}$$

Theorem 3.4. For random variables X and Y, and $a, b \in \mathbb{R}$,

(a) $Var(aX) = a^2 Var(X)$,

(b) $Var(X + Y) = Var(X) + Var(Y)$, if X and Y are uncorrelated.

Proof: (a) Using linearity of $\mathbb{E}$,

$$\begin{aligned}Var(aX) &= \mathbb{E}[(aX - \mathbb{E}[aX])^2] = \mathbb{E}[(aX - a\mathbb{E}[X])^2] = \mathbb{E}[a^2(X - \mathbb{E}[X])^2] \\ &= a^2\mathbb{E}[(X - \mathbb{E}[X])^2] = a^2 Var(X).\end{aligned}$$

(b) Since X and Y are uncorrelated, $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$; and hence

$$\begin{aligned}Var(X + Y) &= \mathbb{E}[(X + Y - \mathbb{E}(X + Y))^2] \\ &= \mathbb{E}[(X - \mathbb{E}(X))^2 + (Y - \mathbb{E}(Y))^2 + 2(XY - \mathbb{E}(X)\mathbb{E}(Y))] \\ &= Var(X) + Var(Y) + 2(\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)) \\ &= Var(X) + Var(Y).\end{aligned}$$

Definition 3.6 (Covariance)**.** The **covariance** of two random variables X and Y is defined as:

$$Cov(X, Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))].$$

The covariance can also be wriiten as:

$$\begin{aligned}Cov(X, Y) &= \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] \\ &= \mathbb{E}[XY - \mathbb{E}[X]Y - \mathbb{E}[Y]X + \mathbb{E}[X]\mathbb{E}[Y]] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

If X and Y are independent, $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(\mathbb{Y})$; so $Cov(X, Y) = 0$.

Theorem 3.5. For random variables X and Y, and $a, b \in \mathbb{R}$,

$$Var(aX + bY) = a^2 Var(X) + 2abCov(X, Y) + b^2 Var(Y).$$

Proof: Using the part (a) and (b) of this theorem, we get

$$\begin{aligned}
Var(aX + bY) &= \mathbb{E}[(aX + bY - \mathbb{E}[aX + bY])^2] = \mathbb{E}[(aX + bY - a\mathbb{E}[X] - b\mathbb{E}[Y])^2] \\
&= \mathbb{E}[a(X - \mathbb{E}[X]) + b(Y - \mathbb{E}[Y])^2] \\
&= \mathbb{E}[a^2(X - \mathbb{E}[X])^2 + b^2(Y - \mathbb{E}[Y])^2 + 2ab(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\
&= a^2\mathbb{E}[(X - \mathbb{E}[X])^2] + b^2\mathbb{E}[(Y - \mathbb{E}[Y])^2] + 2ab\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\
&= a^2 Var(X) + b^2 Var(Y) + 2abCov(X, Y).
\end{aligned}$$

Definition 3.7 (Coorelation)**.** The **correlation coefficient** of random variables X and Y is

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X) \cdot Var(Y)}}$$

as long as the variances are non-zero.

Theorem 3.6. The correlation coefficient ρ satisfies $|\rho(X, Y)| \leq 1$ with equality if and only if $\mathbb{P}(Y = aX + b) = 1$ for some $a, b \in \mathbb{R}$.

Proof: we know that $E\{Y - a\}^2$ is minimized when we choose $a = E(Y)$ so that $E(Y)$ may be interpreted as the best constant predictor of Y. If instead, We choose to predict Y by a linear function of X, say $aX + b$, and measure the error in this prediction by $E\{Y - aX - b\}^2$, then we should choose a and b to minimize this so-called *mean square error*. Clearly, $E(Y - aX - b)^2$ is minimized, for any a,by choosing $b = E(Y - aX) = E(Y) - aE(X)$. With this choice of b, we find a such that

$$\begin{aligned}
E(Y - aX - b)^2 =& E\{(Y - EY) - a(X - EX)\}^2 \\
=& \sigma_Y^2 - 2a\mu_{11} + a^2\sigma_X^2, \qquad \text{where } \sigma_X^2 = Var(X), \sigma_Y^2 = Var(Y), \mu_{11} = Cov(X, Y)
\end{aligned}$$

is minimum. An easy computation shows that the minimum occurs if we choose $a = \frac{\mu_{11}}{\sigma_X^2}$ provided $\sigma_X^2 > 0$. Moreover, we can write

$$\begin{aligned}
\min_{a,b} E(Y - aX - b)^2 =& \min_{a,b} \{\sigma_Y^2 - 2a\mu_{11} + a^2\sigma_X^2\} \\
=& \frac{\sigma_Y^2 - \mu_{11}^2}{\sigma_X^2} \\
=& \sigma_Y^2 \Big\{1 - [\frac{\mu_{11}}{(\sigma_X \sigma_Y)}]^2\Big\}.
\end{aligned}$$

Let us denote

$$\rho = \frac{\mu_{11}}{\sigma_X \sigma_Y}.$$

Since $E(Y - aX - b)^2 \geq 0$,we must have $1 - \rho^2 \geq 0$, or $\rho^2 \leq 1$, or $|\rho| \leq 1$. In the case of equality, we have $|\rho| = 1$, or equivalently, $E(Y - aX - b)^2 = 0$ holds.This implies that $\mathbb{P}(Y = aX + b) = 1$. Here, for $a \neq 0$, we note that the signs of a and ρ are the same. Thus, if $\rho = 1$ then $\mathbb{P}(Y = aX + b)$ when $a > 0$, and if $\rho = -1$, then $a < 0$.

Alternate proof of Cauchy-Schwartz inequality: Random variables X and Y satisfy $\{E(XY)\} \leq E(X^2)E(Y^2)$ with equality holds iff $P(aX = bY) = 1$ for some a and b, at least

one of which is non-zero.
Proof: We know from previous theorem that $|\rho(X,Y)| = |\frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}| \leq 1$, which implies

$$|\mathbb{E}[(X-\mathbb{E}(X))(Y-\mathbb{E}(Y))]| \leq \sqrt{\mathbb{E}[(X-\mathbb{E}[X])^2]\mathbb{E}[(Y-\mathbb{E}[Y])^2]} \tag{3}$$

Let us assume two other random variables U and V with means μ_1, μ_2 and variances σ_1, σ_2, respectively. Now, we consider $X = \frac{U-\mu_1}{\sigma_1}$ and $Y = \frac{V-\mu_2}{\sigma_2}$. Therefore, $E[X] = 0$ and $E[Y] = 0$ and $Var(X) = E[X^2]$ and $Var(Y) = E[Y^2]$. As a result of this assumption, above inequality (3) becomes

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]} \tag{4}$$

Hence, we have the proof.

3.1 Examples of Discrete Random Variables

Bernoulli Random Variable: X is a Bernoulli random variable, if it takes the value 1(success) with probability p and 0(failure) with probability $q = 1-p$. Then

$$f(1) = p,\ f(0) = 1-p,$$

$$\mathbb{E}(X) = \sum_x xf(x) = 1\cdot p + 0\cdot q = p,$$

$$\mathbb{E}(X^2) = \sum_x x^2 f(x) = 1\cdot p + 0\cdot q = p,$$

$$Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = p^2 - p = pq.$$

Binomial Random Variable: The binomial random variable X counts the number of success in n Bernoulli trials. We write it as $X \sim Bin(n,p)$. Then the pmf is given by

$$f(k) = \mathbb{P}(X=k) = \binom{n}{k} p^k q^{n-k},\ \text{where } k = 0,1,2\ldots,n$$

and hence

$$
\begin{aligned}
\mathbb{E}(X) &= \sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k} = \sum_{k=1}^{n}\frac{n!}{(k-1)!(n-k)!}p^k(1-p)^{n-k} \\
&= np\sum_{k=1}^{n}\frac{(n-1)!}{(k-1)!((n-1)-(k-1))!}p^{k-1}(1-p)^{(n-1)-(k-1)} \\
&= np\sum_{k=0}^{n-1}\frac{n!}{k!(n-k)!}p^k(1-p)^{n-1-k} \\
&= np(p+1-p)^{n-1} \\
&= np, \\
\mathbb{E}(X^2) &= \sum_{k=0}^{n} k^2\mathbb{P}(X=k) = \sum_{k=0}^{n}(k+(k-1)k)\mathbb{P}(X=x) \\
&= \sum_{k=0}^{n} k\mathbb{P}(X=k) + \sum_{k=0}^{n} k(k-1)\mathbb{P}(X=k) \\
&= np + \sum_{k=0}^{n} k(k-1)\binom{n}{k}p^k(1-p)^{n-k} \\
&= np + \sum_{k=2}^{n} k(k-1)\frac{n!}{k!(n-k)!}p^k(1-p)^{n-k} \\
&= np + n(n-1)p^2\sum_{k=2}^{n}\frac{(n-2)!}{(k-2)!((n-2)-(k-2))!}p^{k-2}(1-p)^{(n-2)-(k-2)} \\
&= np + n(n-1)p^2\sum_{k=0}^{n-2}\frac{n!}{k!(n-k)!}p^k(1-p)^{(n-2)-k} \\
&= np + n(n-1)p^2(p+1-p)^{n-2} = np + n(n-1)p^2 \\
&= npq + n^2p^2, \\
Var(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = npq.
\end{aligned}
$$

Example 3.1. Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed Bernoulli random variable, then their sum $X = X_1 + X_2 + \ldots + X_n$ is identical to Binomial random variable with parameter (n, p).

$$\mathbb{E}(X) = \mathbb{E}(X_1 + X_2 + \ldots + X_n) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \ldots + \mathbb{E}(X_n) = p + \ldots + p = np,$$

$$Var(X) = Var(X_1 + X_2 + \ldots + X_n) = Var(X_1) + Var(X_2) + \ldots + Var(X_n) = pq + \ldots + pq = npq.$$

Poisson Random Variable: The Poisson random variable X counts the number of success in infinite independent Bernoulli trials. Then for $\lambda > 0$, the pmf is given by

$$f(k) = \frac{\lambda^k}{k!}e^{-\lambda},\ k = 0, 1, 2, \ldots,$$

and hence

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=0}^{\infty} k\frac{e^{-\lambda}\lambda^k}{k!} = \sum_{k=1}^{\infty}\frac{e^{-\lambda}\lambda^k}{(k-1)!} = \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda}e^{\lambda} = \lambda,\\
\mathbb{E}(X^2) &= \sum_{k=0}^{\infty} k^2\mathbb{P}(X=k) = \sum_{k=0}^{\infty}(k+(k-1)k)\mathbb{P}(X=k)\\
&= \sum_{k=0}^{\infty} k\mathbb{P}(X=k) + \sum_{k=0}^{\infty} k(k-1)\mathbb{P}(X=k) = \lambda + \sum_{k=2}^{\infty} k(k-1)\frac{e^{-\lambda}\lambda^k}{k!}\\
&= \lambda + \lambda^2 e^{-\lambda}\sum_{k=2}^{\infty}\frac{\lambda^{k-2}}{(k-2)!} = \lambda + \lambda^2 e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \lambda + \lambda^2\cdot e^{-\lambda}\cdot e^{\lambda}\\
&= \lambda + \lambda^2,\\
Var(X) &= \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.
\end{aligned}$$

Poisson as an approximation to the binomial:
X is a binomial random variable with parameters (n,p), and let $\lambda = np$. Then

$$\begin{aligned}
f(k) = \mathbb{P}(X=k) &= \lim_{\substack{n\to\infty\\ np<\infty}}\binom{n}{k}p^k(1-p)^{n-k} = \lim_{\substack{np=\lambda\\ n\to\infty\\ p\to 0}}\binom{n}{k}p^k(1-p)^{n-k}\\
&= \frac{n(n-1)\dots(n-k+1)}{n^k}\frac{\lambda^k}{k!}\frac{(1-\lambda/n)^n}{(1-\lambda/n)^k}.
\end{aligned}$$

Now, for large n and moderate λ, $(1-\lambda/n)^n \approx e^{-\lambda}$, $(1-\lambda/n)^k \approx 1$, $\frac{n(n-1)\dots(n-k+1)}{n^k} \approx 1$.
Hence,

$$\mathbb{P}(X=k) \approx \frac{e^{-\lambda}}{k!}.$$

Geometric Random Variable: The Geometric random variable counts the number of independent Bernoulli trials required to get the first success. The pmf is given by

$$f(k) = \mathbb{P}(X=k) = (1-p)^{k-1}p,\ k = 1,2,3,\dots$$

and

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=1}^{\infty} k(1-p)^{k-1}p = p\sum_{k=1}^{\infty} k(1-p)^{k-1}\\
&= p\left[\sum_{k=1}^{\infty}(1-p)^{k-1} + \sum_{k=2}^{\infty}(1-p)^{k-1} + \sum_{k=3}^{\infty}(1-p)^{k-1} + \dots\right]\\
&= p\left[\frac{1}{p} + \frac{(1-p)}{p} + \frac{(1-p)^2}{p} + \frac{(1-p)^3}{p} + \dots\right]\\
&= 1 + (1-p) + (1-p)^2 + (1-p)^3 + \dots = \frac{1}{p},\\
\mathbb{E}(X(X-1)) &= \sum_{k=1}^{\infty} k(k-1)p(1-p)^{k-1} = p(1-p)\sum_{k=2}^{\infty} k(k-1)(1-p)^{k-2}\\
&= p(1-p)\sum_{k=2}^{\infty}\frac{d^2}{dp^2}(1-p)^k = p(1-p)\frac{d^2}{dp^2}\sum_{k=2}^{\infty}(1-p)^k\\
&= p(1-p)\frac{d^2}{dp^2}\left(\frac{(1-p)^2}{p}\right) = p(1-p)\frac{2}{p^3} = \frac{2(1-p)}{p^2}.
\end{aligned}$$

The interchange of summation and differentiation is justified by the fact that convergent power series converge uniformly on compact subsets of the set of points where they converge. And hence,

$$\begin{aligned} Var(X) &= \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \mathbb{E}(X(X-1)) + \mathbb{E}(X) - \mathbb{E}(X)^2 \\ &= \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}. \end{aligned}$$

Negative Binomial Random Variable: The Negative random variable X counts the number of Bernoulli trials k required to get r success. Thus, $X = X_1 + X_2 + \ldots + X_r$ where $X_i \overset{D}{\sim} Geom(p)$. Then the pmf is given by

$$f(k) = \mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \ k = r, r+1, \ldots$$

and

$$\mathbb{E}(X) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \ldots + \mathbb{E}(X_r) = \frac{r}{p},$$

$$Var(X) = rVar(X_1) = \frac{r(1-p)}{p^2}.$$

Hyper Geometric Random Variable: The Hyper geometric random variable counts the number of success of drawing red balls from a bag of N balls containing K red balls, in n draws without replacement. Then the pmf will be given by

$$f(k) = \mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \ k = 0, 1, \ldots, K,$$

and

$$\begin{aligned} \mathbb{E}[X] &= \sum_{k=0}^{K} k \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}} = \sum_{k=1}^{K} \frac{k\binom{K}{k}\binom{N-K}{n-k}}{\frac{N}{n}\binom{N-1}{n-1}} = \sum_{k=1}^{K} \frac{K\binom{K-1}{k-1}\binom{N-K}{n-k}}{\frac{N}{n}\binom{N-1}{n-1}} \\ &= \frac{nK}{N} \sum_{k=1}^{K} \frac{\binom{K-1}{k-1}\binom{N-K}{n-k}}{\binom{N-1}{n-1}} = \frac{nK}{N} \sum_{k=0}^{K-1} \frac{\binom{K-1}{k}\binom{(N-1)-(K-1)}{(n-1)-k}}{\binom{N-1}{n-1}}. \end{aligned}$$

The term in summation above is the probability of getting k red balls when $n-1$ balls are drawn from a bag of $N-1$ balls containing $K-1$ red balls. Thus, the sum of these terms over $k = 0, 1, \ldots, K-1$ equals 1. Hence,

$$\mathbb{E}[X] = n\frac{K}{N}.$$

Also,

$$\begin{aligned} \mathbb{E}[X(X-1)] &= \sum_{k=0}^{K} k(k-1) \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}} = \sum_{k=2}^{K} \frac{k\binom{K}{k}\binom{N-K}{n-k}}{\frac{N(N-1)}{n(n-1)}\binom{N-2}{n-2}} \\ &= \sum_{k=2}^{K} \frac{K(K-1)\binom{K-2}{k-2}\binom{N-K}{n-k}}{\frac{N(N-2)}{n(n-2)}\binom{N-2}{n-2}} = \frac{n(n-1)K(K-1)}{N(N-1)} \sum_{k=2}^{K} \frac{\binom{K-2}{k-2}\binom{N-K}{n-k}}{\binom{N-2}{n-2}} \\ &= \frac{n(n-1)K(K-1)}{N(N-1)} \sum_{k=0}^{K-2} \frac{\binom{K-2}{k}\binom{(N-2)-(K-2)}{(n-2)-k}}{\binom{N-2}{n-2}}. \end{aligned}$$

The term in summation above is the probability of getting k red balls when $n-2$ balls are drawn from a bag of $N-2$ balls containing $K-2$ red balls. Thus, the sum of these terms over $k = 0, 1, \ldots, K-2$ equals 1. Hence,

$$\mathbb{E}[X(X-1)] = \frac{n(n-1)K(K-1)}{N(N-1)}.$$

Thus,

$$\begin{aligned} Var(X) &= \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2 \\ &= \frac{n(n-1)K(K-1)}{N(N-1)} + \frac{nK}{N} - \left(\frac{nK}{N}\right)^2 \\ &= \frac{nK(N-n)(N-K)}{N^2(N-1)}. \end{aligned}$$

3.2 Joint Distribution Function

The random variable X and Y on the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ are called **(jointly) discrete** if the random vector (X, Y) takes values in some countable subset of $\mathbb{R}^2$ only.

Definition 3.8 (Joint Distribution Function)**.** The **joint distribution function** $F : \mathbb{R}^2 \to [0, 1]$ of discrete random variables X and Y, is given by

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

Definition 3.9 (Joint Mass function)**.** The joint discrete random variables X, Y have **joint probability mass function** $f : \mathbb{R}^2 \to [0, 1]$ given by

$$f_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

This gives $F_{X,Y}(a, b) = \sum_{x=-\infty}^{a} \sum_{y=-\infty}^{b} f_{X,Y}(x, y)$.

Theorem 3.7. The discrete random variables X and Y are independent if and only if

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \ \forall \ x, y \in \mathbb{R}.$$

Proof: The proof is obvious by definition of independent random variables.

Definition 3.10 (Marginal Distribution Function)**.** Given a joint distribution function $F_{X,Y}$ of random variables X and Y, the **marginal distribution function** of X and Y are given by

$$F_X(x) = \mathbb{P}(X \leq x) = F_{X,Y}(x, \infty),$$

$$F_Y(y) = \mathbb{P}(Y \leq y) = F_{X,Y}(\infty, y).$$

Definition 3.11 (Marginal Mass Function)**.** Given a joint mass function $f_{X,Y}$ of random variables X and Y, the **marginal mass function** of X and Y are given by

$$f_X(x) = \mathbb{P}(X = x) = \sum_y f_{X,Y}(x, y),$$

$$f_Y(y) = \mathbb{P}(Y = y) = \sum_x f_{X,Y}(x, y).$$

This gives $F_X(a) = \sum_{x=-\infty}^{a} \sum_y f_{X,Y}(x, y)$, and $F_Y(b) = \sum_{y=-\infty}^{b} \sum_x f_{X,Y}(x, y)$.

Example 3.2. Let the joint probability mass function of random variable X and Y is

$$f(x, y) = \frac{\alpha^x \beta^y}{x! y!} e^{-\alpha-\beta} \text{ for } x, y = 0, 1, 2, \ldots$$

where $\alpha, \beta > 0$. The marginal mass function of X is

$$f_X(x) = \sum_y f(x, y) = \frac{\alpha^x}{x!} e^{-\alpha} \sum_{y=0}^{\infty} \frac{\beta^y}{y!} e^{-\beta} = \frac{\alpha^x}{x!} e^{-\alpha},$$

and

$$f_Y(y) = \sum_x f(x,y) = \frac{\beta^y}{y!}e^{-\beta}\sum_{y=0}^{\infty}\frac{\alpha^x}{x!}e^{-\alpha} = \frac{\beta^y}{y!}e^{-\beta}.$$

Thus, X has the Poisson distribution with parameter α and Y has the Poisson distribution with parameter β. Clearly $f(x,y) = f_X(x)f_Y(y)$, whence X and Y are independent.

Example 3.3. Let X and Y be two random variables that count the no. of heads and no. tails in N number of independent Bernoulli trials, respectively. Then $X \overset{D}{\sim} Bin(N,p)$ and $Y \overset{D}{\sim} Bin(N, 1-p)$. Thus,

$$\mathbb{P}(X = x) = \binom{N}{x}p^x(1-p)^{N-x}$$

and

$$\mathbb{P}(Y = y) = \binom{N}{y}(1-p)^y p^{N-y}.$$

Also, $\mathbb{P}(X = x, Y = y) = 0$ if $y \neq N - x$; but $\mathbb{P}(X = x), \mathbb{P}(Y = y)$ are never zero. Thus, X and Y are dependent random variables.

3.3 Conditional Distribution Function

Definition 3.12 (Conditional Distribution Function)**.** The **conditional distribution function** of Y given $X = x$, written $F_{Y|X}(\cdot|x)$, is defined by

$$F_{Y|X}(y|x) = \mathbb{P}(Y \leq y|X = x) \ \forall \ x \in \mathbb{R}$$

such that $\mathbb{P}(X = x) > 0$.

Definition 3.13 (Conditional Probability Mass Function)**.** The **conditional probability mass function** of Y given $X = x$, written $f_{Y|X}(\cdot|x)$, is defined by

$$f_{Y|X}(y|x) = \mathbb{P}(Y = y|X = x)$$

for any $x \in \mathbb{R}$ such that $\mathbb{P}(X = x) > 0$.

The conditional probability mass function is also written as

$$f_{Y|X}(y|x) = \mathbb{P}(Y = y|X = x) = \frac{\mathbb{P}(Y = y, X = x)}{\mathbb{P}(X = x)} = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

Definition 3.14 (Conditional Expectation)**.** The **expectation of conditional distribution** of Y given $X = x$, written as $\mathbb{E}(Y|X = x)$, is defined as

$$\mathbb{E}(Y|X = x) = \sum_y y f_{Y|X}(y|x).$$

Let $\psi(x) = \mathbb{E}(Y|X = x)$, then $\psi(X)$ is called the conditional expectation of Y given X, written as $\mathbb{E}(Y|X)$.

Theorem 3.8. The conditional expectation $\psi(X) = \mathbb{E}(Y|X)$ satisfies

$$\mathbb{E}(\mathbb{E}(Y|X)) = \mathbb{E}(Y).$$

Proof:

$$\begin{aligned}\mathbb{E}[\mathbb{E}(Y|X)] &= \mathbb{E}[\sum_y y f_{Y|X}(y|x)] = \sum_x\sum_y y f_{Y|X}(y|x)f_X(x) = \sum_y y\sum_x f_{Y|X}(y|x)f_X(x)\\ &= \sum_y y\sum_x \frac{f_{X,Y}(x,y)}{f_X(x)}f_X(x) = \sum_y y\sum_x f_{X,Y}(x,y) = \sum_y y f_Y(y) = \mathbb{E}[Y].\end{aligned}$$

Example 3.4. A hen lays N eggs, where N has the Poisson distribution with parameter λ. Each egg hateched with probability $p(=1-q)$ independently of the other eggs. Let K be the number of chicks. Find $\mathbb{E}(K|N), K,$ and $N|K$.
$N \overset{D}{\sim} Poisson(\lambda)$ and $K|N \overset{D}{\sim} Binomial(N,p)$; so $\mathbb{P}(N=n) = \frac{e^{-\lambda}\lambda^n}{n!}$ and $\mathbb{P}(K=k|N=n) = \binom{n}{k}p^k(1-p)^{n-k}$.

$$\mathbb{E}(K|N) = Np$$

and

$$\mathbb{E}(K) = \mathbb{E}[\mathbb{E}[K|N]] = \mathbb{E}[Np] = \lambda p.$$

$$\begin{aligned}
f_{N|K}(n|k) = \mathbb{P}(N=n|K=k) &= \frac{\mathbb{P}(N=n, K=k)}{\mathbb{P}(K=k)} = \frac{\mathbb{P}(K=k|N=n)\mathbb{P}(N=n)}{\mathbb{P}(K=k)} \\
&= \frac{\binom{n}{k}p^k(1-p)^{n-k}\frac{e^{-\lambda}\lambda^n}{n!}}{\sum_{n=k}^{\infty}\binom{n}{k}p^k(1-p)^{n-k}\frac{e^{-\lambda}\lambda^n}{n!}} = \frac{\binom{n}{k}p^k(1-p)^{n-k}\frac{e^{-\lambda}\lambda^n}{n!}}{\frac{e^{-\lambda}p^k\lambda^k}{k!}\sum_{n=k}^{\infty}\frac{[(1-p)\lambda]^{n-k}}{(n-k)!}} \\
&= \frac{\binom{n}{k}p^k(1-p)^{n-k}\frac{e^{-\lambda}\lambda^n}{n!}}{\frac{e^{-\lambda}p^k\lambda^k}{k!}e^{\lambda(1-p)}} = \frac{\binom{n}{k}p^k(1-p)^{n-k}\frac{e^{-\lambda}\lambda^n}{n!}}{e^{-\lambda p}\frac{(\lambda p)^k}{k!}} = \frac{e^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!},
\end{aligned}$$

where $n = k, k+1, \ldots$.

$$\begin{aligned}
\mathbb{E}(N|K) &= \sum_{n=k}^{\infty}\frac{ne^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!} = \sum_{n=k}^{\infty}\frac{(n-k+k)e^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!} \\
&= \sum_{n=k+1}^{\infty}\frac{e^{-\lambda q}(\lambda q)^{n-k}}{(n-k-1)!} + k\sum_{n=k}^{\infty}\frac{e^{-\lambda q}(\lambda q)^{n-k}}{(n-k)!} \\
&= \lambda q\sum_{n=k+1}^{\infty}\frac{e^{-\lambda q}(\lambda q)^{n-k-1}}{(n-k-1)!} + k = k + \lambda q.
\end{aligned}$$

Thus, $\mathbb{E}(N|K) = k + \lambda q$.

Theorem 3.9. For random variables X and Y, and $i, j \in \mathbb{N}$,

$$\mathbb{E}[X^iY^j] = \mathbb{E}_X[X^i\mathbb{E}_{Y|X}(Y^j|X)].$$

Proof: The statement can be proved as below,

$$\begin{aligned}
\mathbb{E}[X^iY^j] &= \sum_{i,j}x^iy^jf_{X,Y}(i,j) = \sum_i x^i\Big[\sum_j y^jf_{Y|X}(j|X=i)\Big]f_X(i) \\
&= \sum_i x^i\mathbb{E}_{Y|X}(Y^j|X)f_X(i) = \mathbb{E}_X[X^i\mathbb{E}_{Y|X}[Y^j|X]].
\end{aligned}$$

Theorem 3.10. Prove that, for random variables X and Y,

$$Var(Y) = E(Var(Y|X)) + Var(E(Y|X)).$$

Proof: We know that

$$\begin{aligned}
Var(Y) &= E(Y^2) - [E(Y)]^2, \\
Var(Y|X) &= E(Y^2|X) - [E(Y|X)]^2.
\end{aligned}$$

Then,

$$\begin{aligned}
E[Var(Y|X)] &= E[E(Y^2|X)] - E[\{E(Y|X)\}^2] \\
&= E(Y^2) - E[\{E(Y|X)\}^2] \\
Var[E(Y|X)] &= E[\{E(Y|X)\}^2] - [E[E[(Y|X)]]]^2 \\
&= E[\{E(Y|X)\}^2] - [E(Y)]^2.
\end{aligned}$$

Adding the last two equations, we get

$$Var(Y) = E(Y^2) - [E(Y)]^2 = E(Var(Y|X)) + Var(E(Y|X)).$$

Theorem 3.11 (**Mutual Information**). Let X and Y be discrete random variables with joint mass function f. Show that the **mutual information**

$$I = \mathbb{E}\left[log\left(\frac{f(X,Y)}{f_X(X)f_Y(Y)}\right)\right]$$

satisfies $I \geq 0$, with equality if and only if X and Y are independent.
Proof: We know that $log(y) \leq y - 1$ with equality if and only if $y = 1$. Thus, $-log(y) \geq 1 - y$ implies

$$\begin{aligned}
I &= \mathbb{E}\left[log\left(\frac{f(X,Y)}{f_X(X)f_Y(Y)}\right)\right] = \mathbb{E}\left[-log\left(\frac{f_X(X)f_Y(Y)}{f(X,Y)}\right)\right] \\
&\geq \mathbb{E}\left[1 - \frac{f_X(X)f_Y(Y)}{f(X,Y)}\right] = 1 - \mathbb{E}\left[\frac{f_X(X)f_Y(Y)}{f(X,Y)}\right] \\
&= 1 - \sum_x \sum_y \left(\frac{f_X(x)f_Y(y)}{f(x,y)}\right) f(x,y) = 1 - \sum_x f(x) \sum_y f(y) \\
&= 1 - 1 \cdot 1 = 0.
\end{aligned}$$

Hence $I \geq 0$, and equality holds iff $f(X,Y) = f_X(X)f_Y(Y)$ iff X and Y are independent.

Example 3.5. The lifetime of a machine (in days) is a random variable T with mass function f. Given that that the machine is working after t days, what is the mean subsequent lifetime of the machine when

(a) $f(x) = \frac{1}{N+1},\ x = 0, 1, 2, \ldots, N$

(b) $f(x) = \frac{1}{2^x},\ x = 1, 2, \ldots$

Example 3.6. Let $X_1 \overset{D}{\sim} Geometric(p),\ X_2 \overset{D}{\sim} Geometric(p)$ and X_1, X_2 are independent. Then

$$\begin{aligned}
f_{X_1+X_2}(z) &= \sum_x f_{X_1}(x) f_{X_2}(z-x) = \sum_{x=1}^{z-1} p(1-p)^{x-1} p(1-p)^{z-x-1} \\
&= p^2 \sum_{x=1}^{z-1} (1-p)^{z-2} = \frac{p^2}{(1-p)^2}(1-p)^z \sum_{x=1}^{z-1} 1 \\
&= (z-1)p^2(1-p)^{z-2}.
\end{aligned}$$